

Vision-Based Closed-Loop Autonomous Spacecraft Rendezvous with Robust Relative Pose Estimation

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Abstract—We present a closed-loop simulation framework for vision-based autonomous rendezvous with a tumbling, non-cooperative target, in which perception, estimation and guidance are causally coupled: the camera viewpoint at each step is a consequence of the pose solution at the previous one. The framework combines Hill–Clohessy–Wiltshire relative dynamics, torque-free target attitude propagation, a monocular keypoint front end, RANSAC–EPnP pose estimation with manifold refinement, a multiplicative extended Kalman filter, and LQR and receding-horizon controllers. Each subsystem is validated in isolation — the analytical state transition agrees with DOP853 integration to 4.55×10^{-10} m over three orbits, and RANSAC–EPnP solves every trial up to 30 % outlier contamination where bare EPnP fails in 99 % of trials at 5 %. The closed-loop study identifies measurement *availability*, not pose accuracy, as the binding constraint. Pooled over 3887 control steps of 100 dispersed Monte Carlo trials, the pose solve succeeds on 100 % of steps beyond 6 m and 1.8 % of steps inside 1 m; the transition is predicted in closed form by the field-of-view fill range $z_{fill} = fL/W = 6.25$ m. In a paired noise ablation the median terminal range is unchanged from 0.3 to 3.0 px while the pose error grows tenfold, and the degradation that follows appears in the tail of the distribution — the 95th percentile rises from 0.73 m to 12.47 m and docking success falls from 100 % to 85 % [71, 93] — rather than in the median. The design margin that matters is therefore field of view and keypoint redundancy at short range, not sub-pixel feature localisation.

Index Terms—Spacecraft Rendezvous, Relative Navigation, Computer Vision, EPnP, RANSAC, Pose Estimation, State Estimation, Spacecraft Guidance

I. INTRODUCTION

Autonomous rendezvous and docking (ARD) is a key enabling technology for a wide range of missions including on-orbit servicing, inspection, refueling, assembly, and active debris removal [1], [2]. During a close-proximity operation, the chaser must determine the relative state of the target and use that information to safely guide its own translational and rotational motion towards the docking configuration. Thus, reliable relative navigation is a fundamental prerequisite for autonomous rendezvous and proximity operations (RPO). Vision-based navigation is attractive for such operations because cameras provide rich geometric information at low mass, volume, and power compared with active ranging sensors. Previous studies have demonstrated vision-based relative pose estimation, visual navigation testbeds, and autonomous rendezvous with non-cooperative targets using onboard vision systems [3]–[5].

A challenging aspect for vision-based rendezvous is that errors in the perception subsystem can lead to exponentially large errors in the guidance-and-control (GNC) stages. The image plane measurement noises, target motion, incorrect feature correspondences, and imperfect geometric meshes can introduce significant errors to the system. These errors then affect the controller and degrade the safety and success of the rendezvous maneuver.

For close range pose estimation, the Perspective- n -point (PnP) problem provides a direct means of recovering 6 degrees of freedom relative pose from the known three dimensional target points and their two dimensional image projections. The Efficient PnP method provides an efficient $O(n)$ solution by representing the target points using four virtual control points [6]. However PnP can be sensitive to noises and thus to develop that we are using Random Sample Consensus (RANSAC) which provides a general robust estimation [7]. Nonlinear refinement can then be used to minimise reprojection error and improve initial pose estimation.

The relative translational dynamics of a chaser spacecraft near a target in a near-circular LEO are well described by the Hill–Clohessy–Wiltshire (HCW) linearized equations [8].

This paper presents a modular simulation framework for vision-based autonomous spacecraft rendezvous with a moving and potentially tumbling target. The framework integrates relative orbital dynamics, target attitude dynamics, synthetic camera measurement, pose estimation, nonlinear state estimation, and constrained guidance and control within a single closed-loop environment, so that errors and measurement uncertainties propagate through the navigation and control pipeline exactly as they would onboard.

What distinguishes this framework from the testbeds and benchmarks reviewed in Section II is that the loop is causally closed. Pose-estimation benchmarks score a front end on a fixed image set, so the camera trajectory that produced those images was not generated by a controller acting on the front end’s own output; guidance studies, conversely, assume the state estimate arrives at the control rate. Here the camera viewpoint at step $k + 1$ is a consequence of the pose solution at step k , and a lost measurement therefore propagates. That coupling is what makes the paper’s central quantity observable at all: not how accurate the pose solution is, but how often it exists. Measurement availability is not a free parameter of the perception system — it is a geometric consequence of the

same approach trajectory the controller is generating, and it degrades precisely in the terminal phase, where the control authority available to recover is smallest.

The primary contributions of this work are as follows:

- A closed-loop simulation architecture in which relative dynamics, target attitude motion, synthetic vision, pose estimation, state estimation, and guidance and control are causally coupled, so that measurement dropout is generated by the trajectory rather than imposed on it.
- The identification of measurement *availability*, rather than pose accuracy, as the binding constraint on the closed loop, together with a closed-form criterion — the field-of-view fill range $z_{fill} = fL/W$ — that predicts the range at which availability collapses, validated against the measured knee.
- A dispersed Monte Carlo characterisation, over randomised initial range, bearing, closing speed, tumble rate, target attitude and navigation initialisation error, showing that closed-loop performance is insensitive to pose accuracy across an order of magnitude of image-plane noise and then fails by a threshold mechanism, with the failure attributable to availability rather than accuracy.
- A systematic evaluation of EPnP-based relative pose estimation under increasing image-plane noise, including the planar control-point degeneracy and the effect of refining the control-point coefficients before nonlinear refinement of the pose itself.
- An evaluation of RANSAC-based correspondence rejection under controlled outlier contamination, reporting both pose accuracy and pose-estimation failure rate.
- A numerical validation of the analytical HCW state-transition implementation against high-accuracy DOP853 integration, with a convergence study in solver tolerance, establishing the dynamics as a trustworthy reference for the subsequent closed-loop results.

II. RELATED WORK

This section reviews the bodies of work the present study draws on — autonomous rendezvous, monocular relative pose estimation, robust correspondence estimation, and constrained guidance and control — and then states, in Section II-H, the question their intersection leaves open.

A. Autonomous Rendezvous and Proximity Operations

Automated rendezvous and docking underpins on-orbit servicing, debris removal and in-space assembly [1], [2]; its translational dynamics are conventionally described by the Hill–Clohessy–Wiltshire (HCW) linearisation [8]. The current generation of missions differs in that the target is *non-cooperative*: no docking aids, no retro-reflectors, no telemetry link, and in general tumbling. RemoveDEBRIS exercised a vision-based navigation chain against an uncontrolled target during net and harpoon capture [9], and AVANTI achieved autonomous mid-range rendezvous with a non-cooperative object from angles-only measurements [5]. Both show the guidance layer to be mature, and the binding constraint to have

moved to pose estimates of adequate quality *and adequate frequency* from a passive camera.

B. Model-Based Monocular Relative Pose Estimation

Recovering the six-degree-of-freedom relative pose of a known target from a single image is normally cast as a Perspective- n -Point (PnP) problem, shown viable for rendezvous and docking by Kelsey *et al.* [3] and surveyed across monocular, stereo and LIDAR architectures [10], [11].

The EPnP algorithm [6] remains widely used onboard for its $\mathcal{O}(n)$ complexity, writing the reference points as a weighted combination of four virtual control points and recovering the pose from a compact eigen-decomposition. Two of its properties, treated in Section III-H, matter here: the four-control-point construction degenerates for coplanar reference points, which Lepetit *et al.* handle by dropping to three; and accuracy without the β refinement of the null-space coefficients is materially worse.

Ke and Roumeliotis [12] give a direct algebraic P3P solution avoiding the depth polynomial, and Terzakis and Lourakis [13] present SQPnP, globally optimal for the least-squares PnP cost and stable in planar and quasi-degenerate configurations. The framework here nonetheless retains EPnP, since the question asked — when is a pose solution *available* — is largely independent of the non-iterative solver; Section V-B quantifies the accuracy cost.

C. Learning-Based Pose Estimation and Its Benchmarks

Since 2018, learned front ends have largely displaced hand-designed feature extraction. Sharma *et al.* [14] first applied a convolutional network to non-cooperative spacecraft pose estimation; Sharma and D’Amico [15] developed it into the Spacecraft Pose Network; and Proença and Gao [16] trained a pose regressor on photorealistically rendered scenes, releasing the URSO datasets. The Satellite Pose Estimation Challenge established the SPEED dataset and a common metric [17]; SPEED+ [18] added hardware-in-the-loop and lightbox imagery exposing the synthetic-to-real domain gap, with the follow-on competition analysed by Park *et al.* [19]; and later work attacks that gap through multi-task learning and online refinement [20] and flight-ready architectures trained only on synthetic renderings [21]. Reported accuracies are now a few tenths of a degree in orientation and a fraction of a percent of range in position.

Three features bear on the present paper. First, the benchmarks are *open-loop*: the camera trajectory that produced the scored images was not generated by a controller acting on the network’s own output. Second, the metrics are conditioned on a solution being produced: images on which detection fails are excluded or absorbed into an aggregate score, so the benchmark reports the accuracy of the estimates that exist rather than the rate at which they exist. Third, learned front ends need labelled data and give no geometric guarantee, which is why model-based pipelines remain the reference point for known geometry; the present work follows that route, treating its front end as a source of *intermittent* measurements.

D. Robust Correspondence Estimation

Wrong correspondences degrade pose estimation sharply, and under occlusion, specular glint, Earth-limb background or repeated target structure they are the normal case. Random Sample Consensus [7] remains standard: minimal-sample hypotheses are scored by consensus-set size, with the iteration budget adapted online from the running inlier ratio. LO-RANSAC [22] refits the model on each so-far-best hypothesis's full inlier set; USAC [23] unifies sampling, verification, degeneracy detection and local refinement, with largely complementary component gains; and MAGSAC++ [24] removes the inlier-outlier threshold by marginalising over the noise scale under an iteratively reweighted least-squares optimisation. The framework here uses adaptive-termination RANSAC with EPnP as hypothesis solver and a final inlier refit on the $SO(3)$ manifold; the breakdown point of Section V-C is what a threshold-free estimator would improve.

E. Recursive State Estimation for Tumbling Targets

Kalman-type estimators fuse noisy pose measurements with a relative-motion model. Because an additive extended Kalman filter cannot preserve the unit-quaternion norm through the update, the standard remedy is the multiplicative EKF (MEKF), propagating a small rotation-vector error state composed multiplicatively with a reference quaternion [25]; the unscented Kalman filter [26] instead propagates sigma points and needs no Jacobians, at the cost of more dynamics evaluations. Both are compared here (Section V-D), and, as Section III-J sets out, the attitude residual is formed multiplicatively on the group rather than by differencing global rotation vectors — a distinction invisible for small errors and dominant near the π -radian singularity a tumbling target visits routinely.

The closest work on coupling is that of Pasqualetto Cassinis *et al.* [27], who compare tightly- and loosely-coupled arrangements of a CNN detector, a covariance-aware PnP solver and an MEKF, and show that propagating the image-level covariance into the filter couples rotational and translational states advantageously. That couples the *uncertainty* of the pose solution to the filter, not its *existence* to a controller, because there is no controller in the loop: the trajectory is prescribed.

F. Constrained Guidance and Control for Proximity Operations

Linear quadratic regulation gives a closed-form infinite-horizon feedback law from the discrete algebraic Riccati equation [28], but cannot express the constraints that define a safe approach: a thrust bound, a keep-out zone, an approach corridor. Model predictive control supplies exactly that by re-solving a finite-horizon constrained program at every step [29]. Di Cairano *et al.* [30] formulated proximity manoeuvring with thrust bounds and line-of-sight constraints, showing that for a stationary platform an explicit offline solution removes onboard optimisation entirely; Breger and How [31] generated passively safe trajectories under thruster failure; and Specht *et al.* [32] applied tube-based MPC to rendezvous, guaranteeing constraint satisfaction under bounded disturbance.

The present work does not improve on these formulations. It implements a receding-horizon controller with a second-order approach-cone constraint and soft slack for recursive feasibility (Section III-L) and evaluates it against LQR on the same scenario, keeping the guidance layer representative while the variable of interest remains perception. One point is, however, load-bearing: every formulation above assumes the state estimate is *available at the control rate*. A tube bounds a disturbance, not a measurement outage; an explicit map is a function of a state assumed known. Intermittent measurement is outside the disturbance model these guarantees are built on.

G. Closed-Loop Simulation Frameworks

Gasbarri *et al.* [4] present a ground-based facility for vision-based formation flying, and the hardware-in-the-loop imagery behind SPEED+ and its successors [18], [21] comes from testbeds of the same kind. Such facilities reproduce the photometric realism synthetic rendering struggles with, but not closed-loop causality: the robot follows a commanded trajectory, so the images the estimator sees do not depend on the estimates the controller acted on. The present work makes the opposite trade, forgoing photometric realism for a closed causal loop in which the viewpoint at step $k+1$ follows from the pose solution at step k , so a lost measurement propagates.

H. Gap Addressed by This Work

Taken together, this literature characterises the two halves of a vision-based rendezvous system very well, and characterises them separately.

On one side, the pose-estimation literature reports *accuracy*: rotation and translation error against range, illumination, architecture and domain, on fixed image sets [11], [17], [19]–[21]. These numbers are conditioned on a pose being returned. On the other, the guidance and control literature reports *closed-loop outcomes*: propellant, terminal accuracy, constraint satisfaction, and — in Monte Carlo studies — docking success rate, assuming the state estimate arrives at the control rate [30]–[32]. Work that bridges the two, such as [27], does so at the level of *covariance*: it asks how measurement uncertainty should enter the filter, not how often a measurement enters at all.

The consequence is that a designer holding both halves can predict neither of the two things they most need. Given a measured pose accuracy, no model says whether the closed loop will close; given a closed-loop success rate, there is no attribution of that number to the perception chain, because the two are not connected by any statement about measurement *availability*. Availability is not a free parameter of the perception system: it is a geometric consequence of the same approach trajectory the controller is generating. As the chaser closes, the target's projected extent grows as $1/z$ until it exceeds the field of view, at which point the visible keypoint set collapses — and it collapses precisely in the terminal phase, where the control authority needed to recover is smallest.

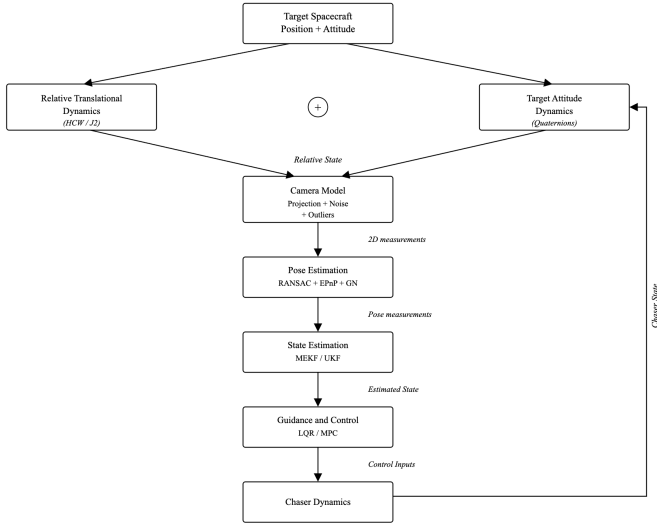


Fig. 1. Closed-loop architecture of the proposed vision-based autonomous spacecraft rendezvous framework.

This paper closes that gap. Its contributions are, first, a closed-loop simulation in which perception, estimation and control are causally coupled, so that measurement dropout is generated by the trajectory rather than imposed on it; second, a closed-form criterion for the range at which field-of-view saturation begins to remove keypoints, $z_{\text{fill}} = fL/W$ (Section V-G), validated against the measured dropout knee; and third, a dispersed Monte Carlo characterisation (Section V-H) showing that closed-loop performance is insensitive to pose accuracy across almost an order of magnitude of pixel noise and then fails abruptly, with the failure attributable to availability rather than accuracy. The practical implication — that the design margin worth buying is field of view and keypoint redundancy at short range, not sub-pixel feature localisation — is not derivable from either half of the literature alone.

III. MATHEMATICAL MODEL

The testbed is a closed-loop simulation of vision-based rendezvous with a tumbling, non-cooperative target, organised into four modules — dynamics, perception, estimation, and guidance-and-control — so that each can be validated in isolation and then exercised inside the complete loop (Fig. 1). At every step the true states are advanced, the target state is rendered into synthetic image measurements, those are reduced to a relative pose, the pose is filtered into a navigation state, and the resulting command acts on the chaser dynamics before the next step. Perception errors therefore propagate into navigation and control, so what is evaluated is the complete autonomous system rather than a set of standalone algorithms.

A. Reference Frames and Relative State

Absolute quantities use the Earth Centred Inertial frame; the relative state uses the target-centred Local Vertical Local Horizontal (LVLH) frame, with radial axis $\hat{\mathbf{x}}$ from the Earth toward the target, $\hat{\mathbf{z}}$ along the target orbital angular momentum and $\hat{\mathbf{y}} = \hat{\mathbf{z}} \times \hat{\mathbf{x}}$. The translational relative state

is $\mathbf{x}_{\text{rel}} = [\mathbf{r}^T \mathbf{v}^T]^T$, and the target attitude is carried as a unit quaternion $\mathbf{q}$ with body-frame angular velocity $\boldsymbol{\omega}$, giving the attitude state $\mathbf{x}_{\text{att}} = [\mathbf{q}^T \boldsymbol{\omega}^T]^T$ subject to $\mathbf{q}^T \mathbf{q} = 1$. The Hamilton convention is used throughout; the dynamics and vision models store $\mathbf{q}$ scalar-first, the estimator scalar-last as $\mathbf{q} = [q_x \ q_y \ q_z \ q_w]^T$.

B. Relative Orbital Dynamics

For a near-circular reference orbit in LEO the relative translational dynamics are the Hill–Clohessy–Wiltshire (HCW) equations [8]. With mean motion $n = \sqrt{\mu/a^3}$, for gravitational parameter μ and target semi-major axis a , the controlled equations are

$$\begin{aligned} \ddot{x} - 2n\dot{y} - 3n^2x &= f_x, \\ \ddot{y} + 2n\dot{x} &= f_y, \\ \ddot{z} + n^2z &= f_z, \end{aligned} \quad (1)$$

where $\mathbf{u} = [f_x \ f_y \ f_z]^T$ is the chaser control acceleration in the LVLH frame. These assemble into the linear time-invariant form $\dot{\mathbf{x}}_{\text{rel}} = \mathbf{A}_{\text{HCW}}\mathbf{x}_{\text{rel}} + \mathbf{B}\mathbf{u}$ with $\mathbf{B} = [\mathbf{0}_{3 \times 3} \ \mathbf{I}_3]^T$, whose closed-form state transition is used by the guidance and control algorithms. A higher-fidelity numerical path is retained for comparison, adding the second-degree zonal harmonic J_2 to the two-body acceleration; it is not exercised by any result reported here, where the unmodelled disturbance is instead injected as a white acceleration on the true trajectory alone.

C. Target Attitude Dynamics

The target is a rigid body in potentially tumbling motion. Its quaternion gives the direction cosine matrix $\mathbf{C}(\mathbf{q})$ from the target body frame to the inertial frame, and the attitude evolves as

$$\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Xi}(\mathbf{q})\boldsymbol{\omega}, \quad \mathbf{I}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (\mathbf{I}\boldsymbol{\omega}), \quad (2)$$

with $\boldsymbol{\Xi}(\mathbf{q})$ the quaternion kinematic matrix, $\mathbf{I}$ the target inertia tensor and $\boldsymbol{\tau}$ the external body-frame torque, zero in the scenarios considered here. Propagation uses an adaptive high-order integrator with the quaternion renormalised each step. For torque-free motion the rotational kinetic energy $\frac{1}{2}\boldsymbol{\omega}^T \mathbf{I} \boldsymbol{\omega}$ and the inertial angular momentum $\mathbf{C}(\mathbf{q})\mathbf{I}\boldsymbol{\omega}$ are invariant, and are monitored as numerical consistency checks.

D. Camera Projection Model

The perception module models a monocular camera on the chaser observing fifteen analytically defined body-frame landmarks $\mathbf{P}_i^b$ on a parametric Ariane 44L upper stage, distributed over the nose, mid-body ring and nozzle, with the body-frame origin at the cylindrical centre and its x -axis along the symmetry axis. The camera sits at the chaser with no mounting rotation or lever arm, so the target attitude supplies the rotation $\mathbf{R}_{bt} \in SO(3)$ and the HCW relative position $\mathbf{r}$ the translation,

$$\mathbf{P}_i^c = \mathbf{R}_{bt}\mathbf{P}_i^b + \mathbf{r}. \quad (3)$$

Landmarks are mapped to the image by the pinhole model, which defines the projection $\pi(\cdot)$,

$$\mathbf{p}_i = \begin{bmatrix} u_i \\ v_i \end{bmatrix} = \begin{bmatrix} f_x X_i^c / Z_i^c + c_x \\ f_y Y_i^c / Z_i^c + c_y \end{bmatrix}, \quad \tilde{\mathbf{p}}_i = \mathbf{p}_i + \boldsymbol{\eta}_i, \quad (4)$$

for focal lengths f_x, f_y and principal point (c_x, c_y) , with zero-mean Gaussian pixel noise $\mathbf{n}_i \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{px}}^2 \mathbf{I}_2)$ added independently to each landmark. The model supports the Brown–Conrady radial and tangential distortion polynomial, disabled here with $k_1 = k_2 = p_1 = p_2 = 0$. A landmark is visible only when $Z_i^c > 0$ and (u_i, v_i) falls inside the $W \times H$ image; the surviving correspondences $\{\tilde{\mathbf{p}}_i \leftrightarrow \mathbf{P}_i^b\}_{i=1}^N$ are the input to the pose-estimation stage. Controlled correspondence outliers can also be injected to test that stage under degraded observations. Camera and noise values are listed in Table I.

E. Perspective- n -Point Pose Estimation

Given N correspondences, the pose-estimation stage recovers $\mathbf{R}_{bt} \in SO(3)$ and $\mathbf{t} \in \mathbb{R}^3$ for which $\mathbf{P}_i^c = \mathbf{R}_{bt} \mathbf{P}_i^b + \mathbf{t}$ reprojects consistently onto the measured image points. The initial estimate uses the Efficient Perspective- n -Point (EPnP) algorithm [6], which requires at least four correspondences, writes each landmark as an affine combination of four virtual control points, and solves the resulting homogeneous system $\mathbf{M}\mathbf{x} = \mathbf{0}$ through the null space of $\mathbf{M}$ by singular value decomposition (Section III-H). Among the candidate null-space solutions the one with the lowest mean reprojection error satisfying the positive-depth chirality constraint is selected, the landmark configuration reconstructed from the barycentric coefficients, and the pose $(\hat{\mathbf{R}}_{bt}, \hat{\mathbf{t}})$ recovered by a rigid-body absolute-orientation solution, whose quality is the reprojection error

$$e_i^{\text{rep}} = \|\tilde{\mathbf{p}}_i - \pi(\hat{\mathbf{R}}_{bt} \mathbf{P}_i^b + \hat{\mathbf{t}})\|_2, \quad \bar{e}_{\text{rep}} = \frac{1}{N} \sum_{i=1}^N e_i^{\text{rep}}. \quad (5)$$

An optional nonlinear refinement of this cost is applied on top of the EPnP solution; because a global rotation-vector parameterisation is singular at $\theta = \pi$, it is formulated on the $SO(3)$ manifold (Section III-I) and is not used in the closed-loop pipeline.

1) *RANSAC-Based Outlier Rejection*: To tolerate incorrect correspondences the EPnP solver is wrapped in a RANSAC stage [7] following the locally optimised strategy of Chum and Matas [22]: a minimal subset is sampled, an EPnP hypothesis generated, and correspondences with $e_i^{\text{rep}} < \tau_{\text{rep}}$ counted as inliers. The largest consensus set wins, with mean inlier reprojection error as the tie-break, and the pose is refitted on its inliers. Sampling terminates adaptively once

$$k \geq \frac{\log(1-p)}{\log(1-w^s)}, \quad (6)$$

where w is the observed inlier ratio, s the minimal sample size and p the desired confidence level, subject to the iteration budget and inlier threshold of Table I.

F. State Estimation

Pose solutions are instantaneous and corrupted by image noise and pose-solver error, so a recursive estimator supplies a temporally consistent relative state. A Multiplicative Extended Kalman Filter (MEKF) [25] and an error-state Unscented Kalman Filter (UKF) [26] are compared, sharing nominal

state, process and measurement models so that the filtering formulation is the only difference. The nominal navigation state is

$$\mathbf{x} = [\mathbf{r}^T \quad \mathbf{v}^T \quad \mathbf{q}^T \quad \boldsymbol{\omega}^T]^T, \quad \delta \mathbf{x} = [\delta \mathbf{r}^T \quad \delta \mathbf{v}^T \quad \delta \boldsymbol{\alpha}^T \quad \delta \boldsymbol{\omega}^T]^T, \quad (7)$$

with $\|\mathbf{q}\| = 1$. Because additive quaternion perturbations violate that constraint, the covariance is carried in the 12-dimensional error state, $\delta \boldsymbol{\alpha}$ being the small-angle attitude error. The nominal state is propagated through the nonlinear dynamics and the covariance through the error-state transition matrix $\mathbf{F}_k$,

$$\mathbf{x}_{k+1}^- = f(\mathbf{x}_k^+, \Delta t), \quad \mathbf{P}_{k+1}^- = \mathbf{F}_k \mathbf{P}_k^+ \mathbf{F}_k^T + \mathbf{Q}_k, \quad (8)$$

where Δt is the sampling interval and $\mathbf{Q}_k$ is acceleration-driven as derived in Section III-K.

The measurement supplied to the estimator is a relative translation and a target attitude. In the filter-isolation experiment it is generated directly from the true state with additive Gaussian noise,

$$\mathbf{z}_k = \begin{bmatrix} \mathbf{r}_k^{\text{true}} \\ \boldsymbol{\rho}_k^{\text{true}} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\nu}_{r,k} \\ \boldsymbol{\nu}_{\rho,k} \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} \sigma_{\text{pos}}^2 \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \sigma_{\text{att}}^2 \mathbf{I}_3 \end{bmatrix}, \quad (9)$$

where $\boldsymbol{\rho}$ is the rotation vector of the target attitude and the noise standard deviations are those of Table I. The complete vision-to-estimation pipeline is evaluated separately in Section V.

1) *Multiplicative Extended Kalman Filter*: With predicted measurement $\hat{\mathbf{z}}_k = h(\mathbf{x}_k^-)$, innovation $\mathbf{y}_k = \mathbf{z}_k - \hat{\mathbf{z}}_k$ and innovation covariance $\mathbf{S}_k = \mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k$, the gain and error-state correction are $\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^T \mathbf{S}_k^{-1}$ and $\delta \mathbf{x}_k = \mathbf{K}_k \mathbf{y}_k$, the attitude component of $\mathbf{y}_k$ being formed multiplicatively on $SO(3)$, which makes $\mathbf{H}_k$ exact and constant (Section III-J). Translational and angular-velocity corrections are applied additively, while the attitude correction is applied multiplicatively to the nominal attitude,

$$\mathbf{q}_k^+ = \text{norm}(\text{Exp}(\delta \boldsymbol{\alpha}_k) \otimes \mathbf{q}_k^-), \quad (10)$$

and the covariance is updated in Joseph form,

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)^T + \mathbf{K}_k \mathbf{R}_k \mathbf{K}_k^T, \quad (11)$$

symmetrised after prediction and update. Measurements statistically inconsistent with the prediction are rejected by a Mahalanobis gate on the squared normalised innovation, $d_k^2 = \mathbf{y}_k^T \mathbf{S}_k^{-1} \mathbf{y}_k \leq \gamma$, with γ the chi-square threshold of Table I; a failing measurement is discarded and the predicted state retained. The attitude part of the innovation is expressed as a principal rotation vector before the distance is evaluated.

2) *Unscented Kalman Filter*: The UKF replaces the analytic linearisation by propagating sigma points through the nonlinear models, using the same 12-dimensional error state. For $n_x = 12$ the scaled unscented transform sets

$$\lambda = \alpha^2(n_x + \kappa) - n_x, \quad W_0^{(m)} = \frac{\lambda}{n_x + \lambda}, \quad (12)$$

so that both the sigma-point spread and the centre weight are controlled by α ; the sigma points and the remaining

weights follow the standard scaled transform [26]. Each error-state sigma point is applied to the nominal state — attitude perturbations multiplicatively, preserving the unit-norm constraint — and propagated through the relative translational and attitude dynamics, giving the cross-covariance $\mathbf{P}_{xz}$ and the innovation covariance $\mathbf{P}_{zz}$, from which $\mathbf{K} = \mathbf{P}_{xz}\mathbf{P}_{zz}^{-1}$ and $\delta\mathbf{x} = \mathbf{K}(\mathbf{z} - \hat{\mathbf{z}})$, the attitude correction again applied multiplicatively. The implemented filter uses $\alpha = 0.5$, $\beta = 2$ and $\kappa = 0$.

G. Guidance and Control

The guidance-and-control module drives the chaser with either a Linear Quadratic Regulator (LQR) or a receding-horizon Model Predictive Controller (MPC), both acting on the HCW relative dynamics. The MPC imposes an explicit thrust constraint and, during the terminal phase, an approach-cone constraint formulated as a softened second-order cone (Section III-L). The commanded acceleration $\mathbf{u}$ updates the chaser state and produces the next camera observation, closing the perception–navigation–control loop.

H. Refinement of the EPnP Control-Point Coefficients

Writing the N world points as affine combinations of control points, $\mathbf{P}_i = \sum_j \alpha_{ij} \mathbf{c}_j$ with $\sum_j \alpha_{ij} = 1$, the projection equations stack into $\mathbf{M}\mathbf{x} = \mathbf{0}$, where $\mathbf{x}$ collects the control points in the camera frame. The solution lies in the null space of $\mathbf{M}$,

$$\mathbf{x} = \sum_{k=1}^n \beta_k \mathbf{v}_k, \quad (13)$$

and the coefficients β_k are fixed by requiring that the inter-control-point distances be preserved:

$$\|\sum_k \beta_k (\mathbf{v}_k^{[i]} - \mathbf{v}_k^{[j]})\|^2 = \|\mathbf{c}_i - \mathbf{c}_j\|^2, \quad i < j. \quad (14)$$

Equation (14) is quadratic in β . Linearising in the products $b_{kl} = \beta_k \beta_l$ gives a linear system $\mathbf{L}\mathbf{b} = \boldsymbol{\rho}$ whose least-squares solution yields an initial β by relinearisation, $\beta_1 = \sqrt{|b_{11}|}$ and $\beta_k = b_{1k}/\beta_1$. This is an *initialisation*, not the answer, and treating it as the answer costs roughly a factor of four in accuracy (Sec. V-B). We refine β by Gauss–Newton directly on (14). With $\mathbf{D}_{ij}$ the matrix whose k -th column is $\mathbf{v}_k^{[i]} - \mathbf{v}_k^{[j]}$ and $\mathbf{d}_{ij} = \mathbf{D}_{ij}\beta$, the residual and its Jacobian are

$$f_{ij}(\beta) = \mathbf{d}_{ij}^\top \mathbf{d}_{ij} - \|\mathbf{c}_i - \mathbf{c}_j\|^2, \quad (15)$$

$$\frac{\partial f_{ij}}{\partial \beta_m} = 2 \mathbf{d}_{ij}^\top \mathbf{D}_{ij} \mathbf{e}_m. \quad (16)$$

Ten iterations of $\beta \leftarrow \beta + \delta$, with δ the least-squares solution of $\mathbf{J}\delta = -\mathbf{f}$, suffice in every case tested.

a) *The coplanar case.*: The control points are the centroid plus the principal axes of the point cloud, scaled by the corresponding singular values. When the cloud is coplanar the third singular value vanishes, so a fourth control point coincides with the centroid, the barycentric system becomes singular, and three columns of $\mathbf{M}$ vanish identically — giving $\mathbf{M}$ a four-dimensional null space for algebraic rather than geometric reasons. The planar configuration must therefore be

detected, from $s_3/s_1 < \epsilon$, and parameterised with *three* control points, so that $\mathbf{x} \in \mathbb{R}^9$ and $\mathbf{M}$ is $2N \times 9$. Approximately 3% of random six-point subsets of the target’s keypoint set are exactly coplanar, so this is not an edge case.

I. Pose Refinement on the $SO(3)$ Manifold

The refinement stage minimises the reprojection cost

$$E(\mathbf{R}, \mathbf{t}) = \sum_i w_i \|\tilde{\mathbf{p}}_i - \pi(\mathbf{R}\mathbf{P}_i + \mathbf{t})\|^2. \quad (17)$$

Parameterising $\mathbf{R} = \exp([\mathbf{r}]_\times)$ by a *global* rotation vector is convenient but singular at $\|\mathbf{r}\| = \pi$, where the derivative of the exponential map is unbounded — not a remote possibility here, since a camera whose boresight is aligned with a world axis produces $\text{tr}(\mathbf{R}) = -1$ exactly, i.e. $\theta = \pi$, and such poses are the natural fixtures for an on-axis approach. We therefore carry $\mathbf{R}$ as a matrix and optimise over a local increment. Using a left-multiplicative update,

$$\mathbf{R} \leftarrow \exp([\delta\boldsymbol{\omega}]_\times) \mathbf{R}, \quad \mathbf{t} \leftarrow \mathbf{t} + \delta\mathbf{t}, \quad (18)$$

and noting that $\exp([\delta]_\times)\mathbf{v} \approx \mathbf{v} - [\mathbf{v}]_\times\delta$, the Jacobian blocks at $\delta = \mathbf{0}$ are

$$\frac{\partial \mathbf{P}_i^c}{\partial \delta\boldsymbol{\omega}} = -[\mathbf{R}\mathbf{P}_i]_\times, \quad \frac{\partial \mathbf{P}_i^c}{\partial \delta\mathbf{t}} = \mathbf{I}_3, \quad (19)$$

so that with the perspective Jacobian

$$\frac{\partial \pi}{\partial \mathbf{P}^c} = \begin{bmatrix} f_x/Z & 0 & -f_x X/Z^2 \\ 0 & f_y/Z & -f_y Y/Z^2 \end{bmatrix} \quad (20)$$

the stacked Jacobian is $\mathbf{J}_i = (\partial\pi/\partial\mathbf{P}^c) [-[\mathbf{R}\mathbf{P}_i]_\times \quad \mathbf{I}_3]$. Because $\delta\boldsymbol{\omega}$ is always near zero, the local parameterisation is well conditioned wherever the optimiser is, and the singularity cannot be reached. Two further conditions keep the iteration honest. The residual and the Jacobian must be evaluated at the same point — clamping Z in one and not the other lets Levenberg–Marquardt accept steps that lower a cost it is not actually computing, and the translation then diverges without bound while the reported cost stays finite. And chirality must be enforced: any pose placing a weighted point at or behind the camera plane is assigned infinite cost and rejected, so failure is visible rather than silent.

J. Multiplicative Attitude Residual

Position enters the residual as a difference in the usual way; attitude cannot, and the reason is worth setting out because the naive form is superficially reasonable. With $\hat{\mathbf{q}}$ the filter’s estimate, encoding both attitudes as global rotation vectors and subtracting, $\boldsymbol{\nu}_{att} = \log(\mathbf{q}_{meas}) - \log(\hat{\mathbf{q}})$, fails at $\theta = \pi$: two attitudes separated by an arbitrarily small rotation, one either side of π , map to nearly antipodal rotation vectors, so a vanishing attitude error produces a residual of magnitude 2π . The linearised residual is then meaningless, and the Jacobian condition number grows without bound as $\theta \rightarrow \pi$. The residual must instead be formed on the group and expressed in the tangent space at the estimate,

$$\delta\boldsymbol{\alpha} = \log(\mathbf{q}_{meas} \otimes \hat{\mathbf{q}}^{-1}), \quad (21)$$

where $\log(\cdot)$ returns the rotation vector of its argument and the sign of the relative quaternion is chosen so that its scalar part is non-negative, selecting the short rotation. This is exact for every attitude, and it has a second benefit. Because the error state is defined by the same left-multiplicative perturbation, $\mathbf{q} = \exp(\frac{1}{2}\delta\boldsymbol{\alpha}) \otimes \hat{\mathbf{q}}$, the residual is the attitude error state to first order, and the measurement Jacobian becomes exact and constant:

$$\mathbf{H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} \end{bmatrix}. \quad (22)$$

No finite differencing is required, and $\mathbf{R}_{att} = \sigma_{att}^2 \mathbf{I}_3$ becomes a genuine tangent-space covariance rather than an approximation. The measurement noise must be injected on the same side for this to hold: a physical attitude perturbation is $\mathbf{q}_{meas} = \exp(\frac{1}{2}\boldsymbol{\eta}) \otimes \mathbf{q}$ with $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma_{att}^2 \mathbf{I}_3)$.

K. Acceleration-Driven Process Noise

The dominant unmodelled effects on the relative translation — differential J_2 , differential drag, thruster execution error — act as an *acceleration*, so the process noise must drive position and velocity jointly. Treating the disturbance as white acceleration of density σ_a over a step Δt gives the standard kinematic block

$$\begin{bmatrix} \mathbf{Q}_{rr} & \mathbf{Q}_{rv} \\ \mathbf{Q}_{vr} & \mathbf{Q}_{vv} \end{bmatrix} = \sigma_a^2 \begin{bmatrix} \frac{1}{3}\Delta t^3 \mathbf{I}_3 & \frac{1}{2}\Delta t^2 \mathbf{I}_3 \\ \frac{1}{2}\Delta t^2 \mathbf{I}_3 & \Delta t \mathbf{I}_3 \end{bmatrix}, \quad (23)$$

with the same structure on the attitude and rate block for an angular acceleration density. The off-diagonal terms are not optional: position and velocity errors driven by a common acceleration are correlated, and a diagonal $\mathbf{Q}$ asserts that they are not, which weakens the information a position measurement carries about velocity. A diagonal alternative that models position as an independent random walk is worse still — a position noise of 0.05 m per step contributes $\mathbf{Q}_{rr} = 2.5 \times 10^{-3} \text{ m}^2$, roughly three hundred times the $8.3 \times 10^{-6} \text{ m}^2$ that a physically plausible $5 \times 10^{-4} \text{ m/s}^2$ disturbance produces over a one-second step. The filter is then needlessly conservative, and its $\mathbf{Q}$ describes a disturbance that is not present in the truth, so the consistency statistics of Eq. (31) become uninformative.

L. Approach Corridor as a Second-Order Cone

The terminal approach is confined to a cone of half-angle θ_c about the $+x$ (radial) approach axis. For a predicted position $\mathbf{r}_k = [x_k, y_k, z_k]^\top$ the constraint is

$$\sqrt{y_k^2 + z_k^2} \leq x_k \tan \theta_c, \quad k = 0, \dots, N-1. \quad (24)$$

Three properties of (24) matter and are each easy to lose. It is a *cone*, not a wedge: both transverse components appear, and constraining $|y_k|$ alone leaves the cross-track axis free. It is *signed*: the right-hand side must be non-negative, so (24) implies $x_k \geq 0$, whereas replacing x_k by $|x_k|$ mirrors the corridor into $x < 0$ and is satisfied by a chaser sitting behind the target. And the constrained quantity is the predicted state $\mathbf{X}(\mathbf{U}) = \mathbf{S}_x \mathbf{e} + \mathbf{S}_u \mathbf{U}$; evaluating the radius on the free response $\mathbf{S}_x \mathbf{e}$ makes the right-hand side independent of

the decision variable, and the constraint ceases to constrain anything. Written for a nonlinear-programming solver, (24) becomes $g_k(\mathbf{U}) \geq 0$ with

$$g_k(\mathbf{U}) = \tan \theta_c x_k(\mathbf{U}) - \sqrt{y_k(\mathbf{U})^2 + z_k(\mathbf{U})^2 + \varepsilon^2}, \quad (25)$$

where ε (here 10^{-6} m) keeps g_k differentiable on the axis at the cost of a marginally conservative constraint. The gradient is

$$\nabla_{\mathbf{U}} g_k = \tan \theta_c \mathbf{S}_u^{x,k} - \frac{y_k \mathbf{S}_u^{y,k} + z_k \mathbf{S}_u^{z,k}}{\sqrt{y_k^2 + z_k^2 + \varepsilon^2}}, \quad (26)$$

with $\mathbf{S}_u^{x,k}$ the row of $\mathbf{S}_u$ selecting x_k .

Imposed as a hard constraint, (24) destroys recursive feasibility: once the chaser lies outside the cone there may be no admissible input returning the first predicted state to it, and the quadratic program is infeasible. Since an infeasible solve is what produces a silent zero-thrust command, we soften the corridor with slack $s_k \geq 0$ under a linear penalty,

$$\min_{\mathbf{U}, \mathbf{s}} \frac{1}{2} \mathbf{U}^\top \mathbf{H} \mathbf{U} + \mathbf{f}^\top \mathbf{U} + \rho \sum_k s_k \quad \text{s.t.} \quad g_k(\mathbf{U}) + s_k \geq 0, \quad (27)$$

which is always feasible and drives $s_k \rightarrow 0$ whenever the hard constraint can be met. The corridor solve is judged on the quantity it exists to reduce, $\sum_k \max(0, -g_k)$, rather than on the solver's own convergence flag, and is attempted only when the box-constrained optimum already violates (24) — an active-set shortcut that leaves the solution unchanged when the corridor is inactive and reduces the per-step cost by two orders of magnitude.

IV. EXPERIMENTAL METHODOLOGY

The architecture is examined in six controlled experiments. Each subsystem is validated against an independent reference before the complete vision–estimation–control loop is exercised, so that numerical and algorithmic errors can be separated from behaviour that emerges only when the components interact. The experiments address three escalating questions: that the constituent models hold, that the individual algorithms tolerate perturbations, and that the assembled architecture still functions when they are coupled.

A. Simulation Environment and Common Parameters

The testbed is implemented in Python. Relative translational motion is propagated in the LVLH frame with the HCW equations and target attitude with the rigid-body model of Section III, using adaptive high-order integration (DOP853) where numerical propagation is required. The target is a parametric Ariane 44L upper stage with fifteen analytically defined body-frame landmarks, projected through the calibrated pinhole camera of Section III-D. Unless an experiment states otherwise, the parameters of Table I apply, and every trial is driven by an explicitly recorded seed so that it is reproducible in isolation.

TABLE I
COMMON SIMULATION AND EXPERIMENTAL PARAMETERS.

Parameter	Value
Camera image size	1024 × 1024 px
Focal length $f_x = f_y$	800 px
Principal point (c_x, c_y)	(512, 512) px
Lens distortion	disabled
Target length	8 m
Target keypoints	15
Minimum keypoints for a solve	6
Nominal pixel noise σ_{px}	1.5 px
RANSAC inlier threshold	2 px
RANSAC iteration budget	500
Pose-failure threshold	15°
Measurement noise (Exp. D)	0.5 m, 0.05 rad
Mahalanobis gate γ	16.812
Filter interval (Exp. D)	10 s
Control interval (Exp. E)	10 s
Control interval (Exp. F)	1 s
MPC prediction horizon	20 steps
Maximum control input (Exp. E)	0.1 m/s ²
Maximum control input (Exp. F)	0.3 m/s ²
Docking criterion	range < 1 m, closing speed < 0.05 m/s

B. Experimental Campaign

a) *A — Dynamics validation:* The closed-form HCW state-transition solution is compared against DOP853 numerical propagation of a periodic trajectory with an initial radial amplitude of 100 m over three orbital periods. The maximum position and velocity discrepancies over the interval are the validation metrics, and the integration tolerance is then swept to confirm that these discrepancies are converging rather than merely small.

b) *B — Pixel-noise sensitivity:* The image-plane noise level is varied and the raw EPnP pose compared with the pose after nonlinear refinement, at a fixed range and target attitude so that the geometry contributes nothing to the trend. This isolates the accuracy the front end can deliver from the accuracy it retains once the refinement stage is removed, and quantifies what that stage is worth.

c) *C — Outlier robustness:* A controlled fraction of the keypoints is replaced by image coordinates drawn uniformly over the frame, leaving the inlier noise fixed. Bare EPnP is compared with RANSAC-EPnP. A trial counts as failed when the solver reports an invalid solution or the recovered rotation error exceeds the threshold in Table I; errors are reported over successful trials and the failure rate over all trials, so that failed solves are not silently removed from the error statistics.

d) *D — State-estimation comparison:* The MEKF and UKF receive an identical sequence of direct pose measurements generated from the true state with additive Gaussian noise, rather than measurements produced by the camera and PnP stages. The comparison therefore characterises the filtering formulation under a known measurement distribution, and is not an end-to-end camera-navigation result. Tracking error, gating behaviour, innovation consistency and wall-clock cost are recorded.

e) *E — Guidance and control comparison:* An infinite-horizon discrete LQR [28] is compared with a receding-

horizon MPC [29] using the LQR solution as its terminal cost, following Breger and How [31]. Both controllers act on the same HCW plant from the same initial state with identical state and control weights, sample time and thrust limit, and the approach-cone constraint is disabled so that thrust saturation is the only active constraint and the difference between them cannot be attributed to constraint feasibility.

f) *F — End-to-end evaluation:* The complete loop is exercised over a dispersed Monte Carlo campaign in which the initial range, bearing, closing speed, tumble rate, target attitude and navigation initialisation error are drawn per trial. Three configurations — perfect state knowledge, direct noisy measurements with the filter, and the full vision pipeline — are compared trial-by-trial on a shared seed sequence, so the configuration is the only difference between paired trials. A second study sweeps the pixel noise over the same paired dispersion to locate the point at which the perception chain becomes limiting, and per-step measurement availability is pooled by instantaneous range to identify the mechanism.

C. Evaluation Metrics

Translation errors are Euclidean norms of the estimate minus the truth. Attitude error is the geodesic angle between the estimated and true orientations,

$$e_R = \cos^{-1} \left[\frac{\text{tr}(\hat{\mathbf{R}}\mathbf{R}^T) - 1}{2} \right], \quad (28)$$

and image-space consistency is measured by the mean reprojection error. For a sequence of N samples the reported RMSE is

$$\text{RMSE}(e) = \sqrt{\frac{1}{N} \sum_{k=1}^N e_k^2}, \quad (29)$$

navigation RMSE being this quantity applied to the position error of the filtered state over a complete closed-loop run. Control effort is the total velocity increment

$$\Delta v = \sum_{k=0}^{N-1} \|\mathbf{u}_k\|_2 \Delta t, \quad (30)$$

and rendezvous accuracy is the terminal range $\|\mathbf{r}_f\|_2$. A trial docks if it satisfies the range and closing-speed criterion of Table I within the simulation horizon; docking success is the fraction of trials that do so. Computational cost is reported as wall-clock time or as time per control step.

V. RESULTS AND DISCUSSION

Experiments A–C verify the individual subsystems against independent references, D and E compare design choices within the estimator and the controller, and F closes the loop and characterises the operating envelope. Every number below is produced by the scripts named in Sec. IV and is reproducible from the stated seeds.

TABLE II
POSE ERROR AGAINST KEYPOINT NOISE. 200 TRIALS PER LEVEL, 15
KEYPOINTS, 20 M RANGE. RMSE OVER SUCCESSFUL SOLVES.

σ_{px} [px]	EPnP		EPnP + LM		solved
	t [cm]	θ [°]	t [cm]	θ [°]	
0.10	0.400	0.056	0.362	0.048	200/200
0.25	1.142	0.143	0.982	0.128	200/200
0.50	2.056	0.275	1.732	0.249	200/200
1.00	4.384	0.567	3.986	0.511	200/200
1.50	6.302	0.865	5.695	0.787	200/200
2.00	7.836	1.095	7.065	1.025	200/200
3.00	13.766	1.696	11.492	1.533	200/200
5.00	23.463	2.885	20.293	2.553	199/200

A. Dynamics Validation

The numerical propagator was validated against the analytical HCW state-transition solution over three orbital periods, using DOP853 with relative and absolute tolerances of 10^{-12} and 10^{-14} . The maximum position discrepancy was 4.55×10^{-10} m and the maximum velocity discrepancy 5.38×10^{-13} m/s. Sweeping the tolerance reduces the maximum position error monotonically from 4.50×10^{-4} m at $(10^{-6}, 10^{-8})$ to 4.73×10^{-11} m at $(10^{-13}, 10^{-15})$: the integrator is converging, not merely reporting small residuals. Two significant digits are quoted because the adaptive DOP853 step sequence depends on the SciPy release, across whose tested versions the result moves by a few per cent. The numerical error is sub-nanometre, nine orders of magnitude below the metre-level separations of interest and eight below the centimetre-level pose errors of Sec. V-B, and is therefore not a contributor to any subsequent result.

B. Vision-Based Pose Estimation

Table II and Fig. 2 report pose sensitivity to image-plane noise at a fixed 20 m range and attitude.

Both errors grow essentially linearly with σ_{px} , as expected with the geometry fixed and the noise small enough for the linearisation about the true pose to hold. At 20 m and $f = 800$ px a 1 px displacement subtends 2.5 cm, consistent with the observed 4.38 cm translation RMSE at $\sigma_{px} = 1$ px given 15 keypoints spanning a 372×112 px image patch.

Levenberg–Marquardt refinement reduces translation error by 9–14% and attitude error by 10–14% over the range tested. We report this modest gain as such because a much larger apparent benefit is easy to obtain accidentally: the closed-form stage of EPnP is accurate only if the control-point coefficients β_k are themselves refined against the inter-control-point distance constraints (Sec. III-H) rather than read directly off the linearised system in the products $\beta_k \beta_l$. Omitting that internal refinement degrades raw EPnP by roughly a factor of four, which then makes the LM stage appear to deliver a large gain. With the reference algorithm implemented in full, most of the achievable accuracy is present before LM runs.

Three cross-checks support the implementation: our EPnP agrees with the OpenCV `SOLVEPNP_EPnP` reference to within a factor of 1.2 in rotation error at $\sigma_{px} = 1$ px; the refined solver matches `cv2.solvePnPRefineLM` to four decimal places at every noise level; and exactly coplanar

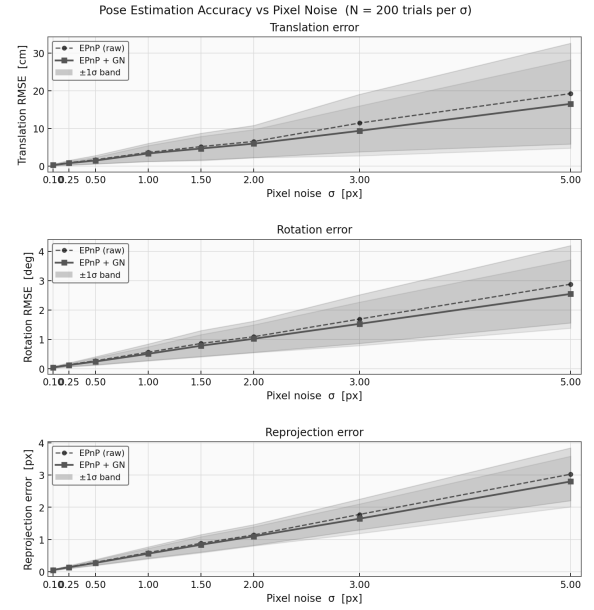


Fig. 2. Pose error against keypoint noise for raw EPnP and after Levenberg–Marquardt refinement. 200 trials per level.

TABLE III
OUTLIER ROBUSTNESS. 200 TRIALS PER LEVEL. ERRORS ARE RMSE
OVER SUCCESSFUL TRIALS; A DASH INDICATES TOO FEW SUCCESSSES TO
REPORT.

f_{out}	EPnP alone			RANSAC–EPnP		
	fail	t [cm]	θ [°]	fail	t [cm]	θ [°]
0%	0%	4.275	0.652	0%	5.175	0.854
5%	99%	14.925	0.792	0%	5.364	0.842
10%	100%	–	–	0%	5.615	0.927
20%	100%	–	–	0%	6.363	0.917
30%	100%	–	–	0%	6.074	1.037

keypoint subsets are solved to 0.0000° , a case the OpenCV EPnP implementation does not handle and one arising in approximately 3% of random six-point samples from this target’s keypoint set.

C. Robustness to Correspondence Outliers

Table III reports the effect of replacing a fraction f_{out} of the 15 keypoints by uniformly distributed image points, with inlier noise held at $\sigma_{px} = 1$ px.

A single gross outlier destroys the least-squares solution: at $f_{out} = 5\%$, fewer than one outlier per frame on average, bare EPnP fails in 99% of trials. Outlier rejection is not a refinement but a precondition for applying a PnP solver to real correspondences. RANSAC–EPnP fails in no trial up to $f_{out} = 30\%$ and degrades only mildly — 5.18 to 6.07 cm in translation, 0.854 to 1.037° in attitude — growth consistent with the reduced number of true inliers available to the final refit rather than with any breakdown of the consensus search.

At $f_{out} = 0\%$ RANSAC is slightly worse than bare EPnP (5.18 cm against 4.275 cm). This is expected rather than a defect: the 2 px inlier gate discards a few legitimate but noisy points, so the final refit uses less data. Robustness carries

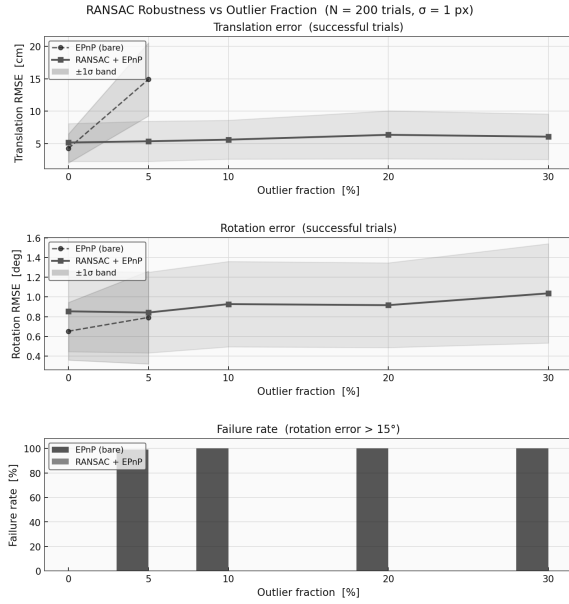


Fig. 3. Pose error and failure rate against outlier fraction, for EPNP alone and for RANSAC-EPNP. 200 trials per level.

TABLE IV
MULTIPLICATIVE EKF AGAINST UKF ON IDENTICAL MEASUREMENTS.
300 STEPS, $\Delta t = 10$ s, SEED 42.

Metric	MEKF	UKF
Position RMSE [m]	0.5997	0.5997
Attitude RMSE [°]	1.2518	1.2802
Velocity RMSE [m/s]	0.0226	0.0226
Max position error [m]	1.5343	1.5343
Max attitude error [°]	7.2282	7.7671
Measurements accepted	300/300	300/300
Wall-clock time [s]	1.602	1.662

a price, and disabling outlier rejection is reasonable when correspondences are known to be clean.

D. Filter Comparison and Consistency

Experiment D isolates the estimator: both filters receive an identical sequence of direct pose measurements over 300 steps, from an initial position offset of 3.9 m and an initial attitude error of 17° , with the perception chain replaced by an equivalent noise model so that filter behaviour is not confounded with pose-solver failures.

The two filters are indistinguishable in translation for a structural rather than an empirical reason: HCW relative translation is exactly linear, so the unscented transform and the analytic Jacobian describe the same map and must agree to machine precision whatever the sigma-point spread. Any difference can appear only through the attitude and rate block, and there the UKF is marginally worse, and marginally *slower* — 1.662 s against 1.602 s, 3.7% — as expected of 25 sigma points per step against a single analytic Jacobian propagation. We therefore find no advantage to the UKF for this problem and use the MEKF throughout the closed-loop study. Two caveats: the sigma-point spread parameter must be $O(0.1-1)$, since with the frequently quoted $\alpha = 10^{-3}$ and $n_x = 12$ one

obtains $n_x + \lambda = 1.2 \times 10^{-5}$, a spread of 0.0035σ and a centre weight of -10^6 , at which point the unscented transform has collapsed onto a finite-difference linearisation and an “EKF versus UKF” comparison compares a filter with itself; and these timings are on unpinned consumer hardware, so the 3.7% gap should be read as “no meaningful difference”, not as a measurement.

Filter consistency is assessed through the normalised innovation squared,

$$\text{NIS}_k = \mathbf{v}_k^\top \mathbf{S}_k^{-1} \mathbf{v}_k, \quad (31)$$

which for a consistent filter is χ^2 distributed with $n_z = 6$ degrees of freedom. Over 3600 updates (12 seeds $\times$ 300 steps) the sample mean is $\bar{\text{NIS}} = 6.044$, against a 95% confidence interval on the mean of $[5.887, 6.113]$ obtained from $n_z \pm 1.96\sqrt{2n_z/N}$ using $\text{Var}[\chi_k^2] = 2k$; the fraction of updates exceeding $\chi_{6,0.99}^2 = 16.81$ is 0.97%, against a nominal 1%. The filter is therefore consistent: its reported covariance matches the errors it actually makes, and the Mahalanobis gate rejects at the designed rate.

This is a stronger statement than a small RMSE, and it rests on three modelling choices that are easy to get wrong. The attitude residual must be formed multiplicatively on $SO(3)$ (Sec. III-J); the process noise must be acceleration-driven with the position-velocity cross terms retained (Sec. III-K); and the truth must carry a disturbance that the filter’s $\mathbf{Q}$ actually describes. Substituting a global rotation-vector residual yields $\bar{\text{NIS}} = 6.82$, an over-confident filter; removing the truth disturbance yields 5.77, a conservative one.

E. Guidance: LQR and MPC

Experiment E compares the infinite-horizon LQR with the receding-horizon MPC on a 50 m along-track approach, sharing weights, sample time and thrust limit, with the LQR solution as the MPC terminal cost and the corridor disabled so that thrust saturation is the only active constraint.

MPC reaches the same terminal accuracy for 5.00 m/s against 16.83 m/s, a 70% reduction in propellant. The mechanism must be stated precisely, because the comparison is easy to over-read. With shared weights and the LQR terminal cost, an MPC whose constraints are inactive reproduces the infinite-horizon LQR exactly — verified here to 4.8×10^{-7} . The entire difference is therefore constraint handling, not prediction: the LQR gain demands 1.77 m/s^2 at the initial condition, seventeen times the actuator limit, and is then clipped per axis, whereas MPC distributes the available authority over its horizon. The honest framing is *MPC against a saturated LQR*, and the result is a statement about anti-windup; a constrained-LQR or reference-governor baseline would narrow the gap and is the appropriate comparison for future work.

The measured 0.57 ms per step reflects an active-set shortcut: the corridor-constrained problem is solved only when the corridor is violated at the box-constrained optimum. With the corridor active throughout, the per-step cost rises by two orders of magnitude and is dominated by a few hard early steps, for which a median and 95th percentile are the appropriate statistics.

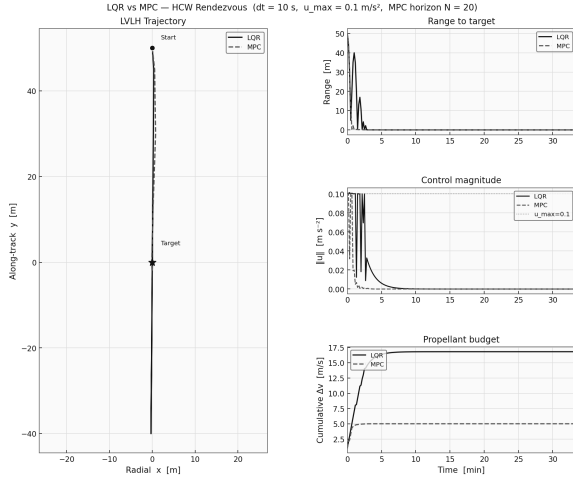


Fig. 4. LQR and MPC approach trajectories in the LVLH plane.

TABLE V
CLOSED-LOOP PERFORMANCE OVER 12 SEEDS. DOCKING IS RANGE
< 1 M WITH CLOSING SPEED < 0.05 M/S.

Metric	Mean	Std
Docking step	42.7	0.6
Δv to docking [m/s]	2.562	0.052
Navigation RMSE [m]	0.345	0.017
Terminal range [m]	0.427	0.169
Peak closing speed [m/s]	1.332	0.027
Vision dropout [%]	36.3	0.8
Thrust-saturated steps	0	0

F. Closed-Loop Performance

The full pipeline was run from a 30 m initial separation with a tumbling target ($\omega_0 = [0.02, 0.05, 0.01]$ rad/s). Table V summarises 12 seeds: all dock, in 2.56 m/s of Δv , with no step reaching the thrust limit.

The absence of saturation is a design requirement rather than an observation: an LQR gain tuned without regard to the actuator limit demands approximately ninety times u_{max} at 30 m range, saturates on a quarter of all steps, arrives at 4 m/s and passes through the target before returning — a limit cycle rather than a rendezvous. Weighting control so that the unsaturated demand respects u_{max} , and commanding the controller to a docking-port hold point on the approach axis rather than to the target centre of mass, reduces the propellant requirement by a factor of sixteen and the peak closing speed by a factor of three.

G. Measurement Availability as the Limiting Factor

The principal finding of the closed-loop study is that the binding constraint on this architecture is not pose accuracy but *measurement availability*, and that availability is governed by geometry rather than by noise.

Availability must be measured against the range the camera is *at*, not against the range a trial started from. Every trial that closes traverses the whole range interval, so a trial-level dropout rate averages the far field, where every keypoint projects inside the frame, with the terminal phase, where the target overfills the sensor. That rate is accordingly almost

TABLE VI

MEASUREMENT AVAILABILITY AGAINST INSTANTANEOUS RANGE, POOLED OVER 3887 CONTROL STEPS OF 100 DISPERSED MONTE CARLO TRIALS AT $\sigma_{px} = 1.5$ px. A SOLVE IS ATTEMPTED ONLY WHEN AT LEAST SIX KEYPOINTS PASS THE VISIBILITY GATE. INTERVALS ARE WILSON 95 % SCORE INTERVALS ON THE PER-STEP AVAILABILITY.

Range [m]	Steps	Mean visible	Availability	95 % CI
10–50	1686	15.00	100.0%	[99.8, 100.0]
8–10	201	14.44	100.0%	[98.1, 100.0]
6–8	234	10.93	100.0%	[98.4, 100.0]
5–6	131	8.63	96.9%	[92.4, 98.8]
4–5	156	7.83	85.3%	[78.8, 90.0]
3–4	167	7.07	67.1%	[59.6, 73.7]
2–3	212	4.83	26.9%	[21.4, 33.2]
1–2	318	3.42	9.7%	[7.0, 13.5]
< 1	782	2.51	1.8%	[1.1, 3.0]

constant across the dispersion: 34.3%, 33.3% and 31.7% for trials beginning in the 10–20 m, 20–30 m and 30–50 m bands respectively, which says nothing at all. Table VI therefore pools the per-step outcome of every step of every trial in the dispersed campaign of Section V-H and bins it by instantaneous range.

The profile is a step, not a gradient: availability is statistically indistinguishable from unity at every range beyond 6 m and falls by a factor of fifty-five between the 5–6 m and the sub-metre bins.

The mechanism is field-of-view exit, and it is predictable in closed form. A target of characteristic length L fills a sensor of width W pixels at a range

$$z_{fill} = \frac{fL}{W}, \quad (32)$$

which for the 8 m Ariane upper stage with $f = 800$ px and $W = 1024$ px gives $z_{fill} = 6.25$ m. The first bin in Table VI whose availability interval excludes unity is 5–6 m, immediately inside that prediction. Inside z_{fill} the number of keypoints simultaneously within the image falls below the six required for a minimal RANSAC sample — the mean visible count crosses six between the 4–5 m and 3–4 m bins — and the pose solution becomes unavailable regardless of detector accuracy.

This has a direct consequence for the noise ablation. Table VII sweeps σ_{px} over 40 dispersed trials per level, *paired*: the same seed sequence generates the initial position, velocity, tumble rate and attitude at every noise level, so the level-to-level difference is the effect of the parameter and not of a fresh draw of initial conditions.

The median terminal range is flat from 0.3 to 3.0 px — it varies by 4 cm while the noise grows tenfold — and the degradation appears first in the *tail*, not in the centre of the distribution. The 95th percentile rises from 0.73 m at 2 px to 8.22 m at 3 px and 12.47 m at 4 px, and docking success falls from 100% to 85% over the same interval. The median trial at 4 px docks as tightly as the median trial at 0.3 px while roughly one trial in seven diverges completely: the signature of a threshold, not of drift. The threshold is the point at which the pose solver’s success rate inside z_{fill} falls low enough that the filter coasts on prediction for the whole terminal phase,

TABLE VII

TERMINAL ACCURACY AND DOCKING SUCCESS AGAINST KEYPOINT NOISE. 40 DISPERSED TRIALS PER LEVEL, PAIRED ACROSS LEVELS, FULL VISION–MEKF PIPELINE. TERMINAL RANGE IS REPORTED AS THE MEDIAN WITH 5TH AND 95TH PERCENTILES: A SINGLE DIVERGING TRIAL DOMINATES THE MEAN AND ITS STANDARD DEVIATION. DOCKING INTERVALS ARE WILSON 95 % SCORE INTERVALS.

σ_{px} [px]	Terminal range [m]	Docking	95 % CI
0.3	0.34 [0.11, 0.78]	100%	[91, 100]
0.7	0.33 [0.14, 0.82]	100%	[91, 100]
1.0	0.32 [0.11, 0.81]	100%	[91, 100]
1.5	0.30 [0.13, 0.82]	100%	[91, 100]
2.0	0.32 [0.09, 0.73]	98%	[87, 100]
3.0	0.34 [0.15, 8.22]	95%	[83, 99]
4.0	0.51 [0.17, 12.47]	85%	[71, 93]

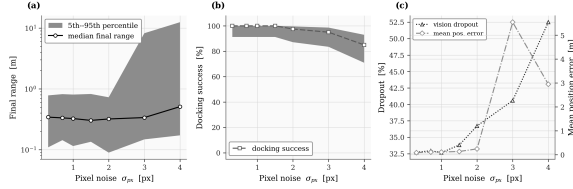


Fig. 5. Terminal range, docking success and vision dropout against keypoint noise, 40 paired dispersed trials per level. The envelope is flat to 2 px; the degradation beyond it appears in the tail of the terminal range distribution while the median is unchanged.

and the accumulated dead-reckoning error exceeds the docking corridor before the range gate is satisfied.

The dispersion also weakens a conclusion that a single-trajectory study would have supported. Repeating this ablation along one fixed approach trajectory, varying only the noise realisation, yields a far sharper collapse — 65% success at 4 px with a terminal-range standard deviation above 100 m — because every trial then encounters the field-of-view knee at the same geometry and the same phase of the tumble. Dispersing the initial condition spreads that encounter, and the honest statement of the effect is the milder one in Table VII. We report both because the difference is itself the argument for dispersing: a Monte Carlo campaign that varies only the random seed measures the repeatability of one trajectory, not the robustness of a design.

Two implications follow. First, the useful figure of merit for a monocular front end in this role is not pose RMSE but the probability of returning a usable solution, and the two do not degrade together: between 0.3 and 2.0 px the pose error grows by a factor of seven (Table II) while closed-loop performance is unchanged. Second, terminal approach inside z_{fill} requires a change of sensing modality — a wider lens, a second short-range camera, or a close-range keypoint subset chosen to remain in frame — and no improvement in the pose solver will substitute for it. The design margin worth buying is field of view and keypoint redundancy at short range, not sub-pixel feature localisation.

H. Dispersed Monte Carlo Campaign

The configurations are compared over a dispersed Monte Carlo campaign rather than over repeated runs of one trajectory. Each trial draws an initial range log-uniformly on

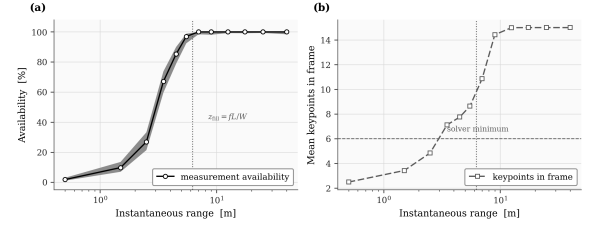


Fig. 6. (a) Per-step measurement availability against instantaneous range, pooled over the 100-trial dispersed campaign, with the closed-form fill range $z_{fill} = fL/W = 6.25$ m marked. Shading is the Wilson 95 % interval. (b) Mean number of keypoints passing the visibility gate; the six-keypoint solver minimum is crossed just inside z_{fill} .

TABLE VIII

CONFIGURATION COMPARISON OVER 100 DISPERSED TRIALS EACH, PAIRED ACROSS CONFIGURATIONS, $\sigma_{px} = 1.5$ PX WHERE APPLICABLE. TERMINAL RANGE IS THE MEDIAN WITH 5TH AND 95TH PERCENTILES; DOCKING INTERVALS ARE WILSON 95 % SCORE INTERVALS.

Configuration	Range [m]	Δv [m/s]	Docking
Perfect state	0.33	2.53 ± 0.80	100% [96, 100]
Direct measurement + MEKF	0.54	2.65 ± 0.81	100% [96, 100]
Full vision + MEKF	0.33	2.52 ± 0.80	100% [96, 100]

[15, 45] m, a bearing uniformly within a 20° cone about the radial approach axis, a closing speed uniformly on $[-0.15, -0.02]$ m/s with a 1 cm/s transverse component, a tumble rate uniformly on $[0.005, 0.09]$ rad/s about an isotropically distributed axis, an initial target attitude uniform on $SO(3)$ by Shoemake’s method, and a navigation initialisation error of 1.5 m and 2 cm/s per axis (1σ). Trials are seeded from a single *SeedSequence*, spawned per trial and split again into an independent dispersion stream and simulator stream, so any trial is reproducible in isolation, is independent of the campaign size and of execution order, and is identical whether the campaign runs serially or in parallel. The three configurations share that seed sequence and are therefore compared trial-by-trial on identical initial conditions.

The three configurations lie within 5% of one another in propellant and dock in every one of the 300 trials. We draw a deliberately narrow conclusion: at $\sigma_{px} = 1.5$ px the perception chain is *not* the limiting subsystem, and closed-loop performance is set by the guidance law and the actuator limit. It would be a mistake to read the full pipeline’s marginal advantage over the direct-measurement baseline as evidence that vision outperforms a half-metre position sensor: the two configurations have different measurement error magnitudes, and the comparison is informative only about which subsystem is binding — which at this noise level is neither. It becomes discriminating only above 3 px, where Table VII shows the perception chain taking over as the limiting factor.

Two aspects of the campaign design are worth stating explicitly, because both were defects in an earlier version of this study. First, the perfect-state configuration has no sensor noise anywhere, so if only the random seed were varied its trials would be bit-identical and its reported distribution would be an artefact; dispersing the initial condition gives it a genuine distribution, and the Δv spread of ± 0.80 m/s in Table VIII

is the spread induced by the initial condition alone, which is the correct baseline against which to read the other two rows. Second, mean Δv over all trials is not a propellant budget when some trials fail, so the campaign reports it conditioned on docking as well; here the two coincide because every trial docked, but they diverge in the ablation of Table VII.

VI. CONCLUSION

This paper presented a modular closed-loop simulation framework for vision-based rendezvous with a tumbling non-cooperative target, and asked what the pose-estimation and guidance literatures do not answer between them: given a monocular front end of known accuracy, will the loop close?

Each subsystem was validated in isolation. The analytical HCW state-transition implementation agrees with DOP853 integration to 4.55×10^{-10} m over three orbital periods. EPnP with full β refinement matches a reference implementation to within a factor of 1.2 in rotation error and solves exactly coplanar keypoint subsets — approximately 3 % of random six-point samples — that the reference rejects. RANSAC–EPnP fails in no trial up to 30 % outlier contamination; bare EPnP fails in 99 % of trials at 5 %. The MEKF and UKF are indistinguishable in translation, as exactly linear HCW dynamics require, and differ only in attitude, where the MEKF is marginally better and faster.

The closed-loop result is that the binding constraint is measurement *availability*, set by geometry rather than noise. Pooled over 3887 control steps of 100 dispersed Monte Carlo trials, the pose solve succeeds on 100 % of steps beyond 6 m and on 1.8 % inside 1 m. The transition is predictable in closed form: a target of characteristic length L fills a sensor of width W pixels at $z_{fill} = fL/W$, here 6.25 m, and the first range bin whose availability interval excludes unity is 5–6 m. Inside z_{fill} the keypoints simultaneously in frame fall below the six a minimal RANSAC sample requires, and no better solver recovers them.

Pose RMSE is therefore the wrong figure of merit here. In a paired ablation over 40 dispersed trials per level, the median terminal range is flat from 0.3 to 3.0 px, varying by 4 cm while keypoint noise grows tenfold and the pose error with it, and the degradation that follows is a threshold rather than a drift: the median trial at 4 px docks as tightly as the median trial at 0.3 px, while the 95th percentile of terminal range rises from 0.73 m to 12.47 m and docking success falls from 100 % to 85 % [71, 93]. Roughly one trial in seven diverges completely, because the solver’s success rate inside z_{fill} falls low enough that the filter coasts on prediction through the terminal phase. Terminal approach inside the fill range thus requires a change of sensing — a wider lens, a second short-range camera, or a keypoint subset chosen to stay in frame — and that, not sub-pixel feature localisation, is the margin worth buying.

Two methodological points support these conclusions. The campaign disperses initial range, bearing, closing speed, tumble rate, target attitude and navigation initialisation error, not only the seed; along a single fixed approach the same ablation collapses far more sharply, because every trial then meets the

field-of-view knee at identical geometry and tumble phase. Availability must also be measured against the range the camera is at, not the range a trial began from: binned by initial range the dropout rate is 34.3 %, 33.3 % and 31.7 %, carrying no information because every trial that closes traverses the whole interval.

The principal limitation is the front end. Correspondences are known and their projection analytic, so the perception chain is idealised: real imagery adds illumination variation, specular glint, Earth-limb background, partial visibility and matching ambiguity, all of which would reduce availability further and none of which would move z_{fill} , set by the optics and target alone. The availability result is therefore best read as an upper bound. Future work should re-measure the availability profile against a rendered or hardware-in-the-loop front end; extend the analysis to a threshold-free robust estimator, whose principal benefit would be to the breakdown point measured here; and address the terminal phase directly, through the multi-sensor arrangement the fill criterion implies or a guidance law that anticipates the outage rather than coasting through it.

DATA AND CODE AVAILABILITY

The simulation framework, the experiment scripts that generate every figure and table in this paper, and the per-trial Monte Carlo records are available from the author on reasonable request. All results are reproducible from the released code: each Monte Carlo trial is seeded from a single spawned seed sequence and is therefore reproducible in isolation, independent of the campaign size, of execution order, and of whether the campaign is run serially or in parallel.

USE OF GENERATIVE AI

The author used an AI coding assistant for code review, defect diagnosis and manuscript editing. All algorithms, experiments and results were specified, executed and verified by the author, who takes full responsibility for the content of this work.

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